

Gold Future/ETF Analysis

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Table 1: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.98218	-9.048×10^{-5}	0.92459	0.00269
2-Month	0.98709	-1.255×10^{-4}	0.92244	0.00275
6-Month	0.98870	-9.035×10^{-5}	0.91718	0.00285
12-Month	0.92794	9.349×10^{-5}	0.76020	0.00520

Table 2: A summary of the slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.96096	0.96652	0.96792	0.91183
	5	0.82649	0.82903	0.86076	0.93489
	10	0.81291	0.81483	0.84672	0.93109
	15	0.81350	0.81488	0.84904	0.93986
R^2	1	0.90794	0.90739	0.90188	0.75375
	5	0.79750	0.79703	0.83189	0.83831
	10	0.80108	0.80008	0.83108	0.85175
	15	0.80114	0.80018	0.83230	0.85757

Table 3: A summary of the weights and in/out of sample RMSEs for portfolios of 1, 2, and 3 futures contracts.

Portfolio		w_0	w_1	w_2	w_3	RMSE (in)	RMSE (out)
1 Futures	1-m	-0.16053	1.16053			0.00285	0.00427
	2-m	-0.17993	1.17993			0.00299	0.00450
	6-m	-0.18732	1.18732			0.00295	0.00541
	12-m	-0.06259	1.06259			0.00483	0.00684
2 Futures	1-m, 2-m	-0.16936	0.85695	0.31241		0.00283	0.00434
	1-m, 6-m	-0.17736	0.71410	0.46326		0.00279	0.00450
	1-m, 12-m	-0.19057	1.01853	0.17204		0.00279	0.00445
	2-m, 6-m	-0.19181	0.53762	0.65419		0.00285	0.00478
	2-m, 12-m	-0.20928	1.01944	0.18983		0.00291	0.00473
	6-m, 12-m	-0.21749	1.04815	0.16933		0.00287	0.00504
3 Futures	1-m, 2-m, 6-m	-0.18008	0.60862	0.14966	0.42180	0.00278	0.00450
	1-m, 2-m, 12-m	-0.19688	0.79086	0.24270	0.16333	0.00278	0.00450
	1-m, 6-m, 12-m	-0.20496	0.64458	0.41532	0.14506	0.00274	0.00464
	2-m, 6-m, 12-m	-0.21907	0.47364	0.59770	0.14773	0.00280	0.00488